

OFFICIAL RULEBOOK, PROBLEM STATEMENTS & PITCH GUIDELINES

Event Name & Theme	AdhigamAI Inter-College Hackathon AI & Technology for Real-World Impact
Competition Structure	Round 1: Online PPT Idea Submission Round 2: Physical Pitch & Working Prototype Demo
Grand Finale Venue	AdhigamAI Master Hub - Sri Sukhmani College of Technology & Management, Dera Bassi, Punjab
Eligibility & Team Size	Open to 1st Year to Final Year students (All Branches / Courses) Teams of 3 to 5 Members (Solo Not Allowed)
Tracks & Challenges	15 Problem Statements Across 5 Domains: Farmers, Army/Defence, Startups, Real-Life, Road Safety
Registration Link & Fee	https://adhigamai.com/hackathon2026 Free Participation (No Registration Fee)

1. Event Overview & Competition Philosophy

The AdhigamAI Inter-College Hackathon is a premier multi-disciplinary innovation competition conceived to bridge the gap between academic classroom learning, advanced Artificial Intelligence (AI) technologies, and pragmatic grassroots problem-solving. It provides a high-stakes, rigorous platform for ambitious students to formulate, architect, and demonstrate tangible technology solutions that address acute real-world challenges faced by farmers, national defence personnel, emerging startup ecosystems, everyday citizens, and road safety infrastructure.

The hackathon operates under a structured Two-Round Framework that balances wide accessibility with deep technical rigor:

- **Phase 1 Online Idea Qualifier:** Teams select one designated Problem Statement from the official repository of 15 challenges across 5 domains, formulate a robust technological/AI solution, and submit an exhaustive 10-slide PowerPoint presentation (PPT) via the official submission portal for evaluation by an expert screening jury.
- **Phase 2 Offline Pitch Round (Grand Finale):** The top shortlisted finalist teams from Round 1 advance to the grand offline finale hosted at the AdhigamAI Master Hub (Sri Sukhmani College, Dera Bassi). Finalists present their strategic solution through a formal pitch presentation, demonstrate a functioning working prototype / MVP live on stage, and defend their architecture during an intensive Q&A session with industry leaders and technical juries.

Why the Pitch Round Model Traditional hackathons frequently exhaust participants through grueling 24-to-36-hour sprint coding on site, often resulting in hurried, half-baked code and sleep-deprived presentations. The AdhigamAI Hackathon employs an advanced 'Pitch Round' model: teams build, refine, and test their working prototypes thoroughly ahead of the finale, ensuring that the offline event is entirely dedicated to executive pitch delivery, live architectural validation, functional prototype demonstration, and direct high-value networking with mentors, investors, and campus chapters.

2. Master Structure & Hackathon Lifecycle

The following master workflow represents the end-to-end operational roadmap of the AdhigamAI Inter-College Hackathon, adapted directly from the Master Architecture schema:

Phase / Round	Mode	Stage / Activity	Key Actions & Responsibilities	Core Deliverable / Output
Phase 1: Registration	Online	Team Formation & Registration	Students form teams of 3-5 members, nominate Team Leader, and register via the official form.	Verified Team Registrations
Phase 1: Orientation	Online	Participant Orientation Webinar	AdhigamAI briefing on theme, problem statement tracks, PPT blueprint, and judging expectations.	Orientation session completed
Round 1: Preparation	Online	Problem Selection & Ideation	Teams choose 1 PS, formulate their AI/engineering approach, target user journey, and build plan.	Finalized Solution Blueprint
Round 1: Submission	Online	Idea Deck + PPT Submission	Submit project title, problem summary, tech stack, and 10-slide PPT via official portal.	Complete Round 1 Submission
Round 1: Screening	Internal	Eligibility & Verification Review	Verification of enrollment credentials, team composition, completeness, and originality screening.	Screened Submission Pool
Round 1: Evaluation	Online	Expert Jury Review & Scoring	Technical and domain jury evaluates submissions based on 6 core criteria (0-100 scale).	Jury Scorecards & Rankings
Qualification	Online	Shortlisting & Announcement	Top-scoring teams selected as Finalists and formally notified via email and portal dashboard.	Official Finalist List
Qualification	Online	Finalist Confirmation & Logistics	Finalists confirm participation, receive venue access passes, reporting rules, and prototype criteria.	Finalist Attendance Confirmed

Phase / Round	Mode	Stage / Activity	Key Actions & Responsibilities	Core Deliverable / Output
Round 2: Pitching	Offline	Pitch & Prototype Demo Sessions	Sessions 1 & 2: 10 12 min slot per team (PPT pitch + live working prototype demo + jury Q&A).	Round 2 Jury Score Sheets
Round 2: Deliberation	Offline	Jury Deliberation & Scoring	Compiling scores across all criteria, normalising results, and determining podium winners.	Final Winner Rankings
Closing & Awards	Offline	Valedictory & Prize Distribution	Jury remarks, announcement of Winners, Runner-ups, trophy presentation, and certificates.	Trophies, Grants & Certificates

3. Eligibility, Team Formation & Registration Protocol

3.1 Eligibility Criteria

- **Academic Standing:** Open to all bonafide undergraduate and postgraduate students enrolled in recognized universities and colleges, from 1st Year to Final Year.
- **Academic Branches:** Students from all branches and academic streams are eligible including Computer Science, IT, Electronics, Mechanical, Civil, Electrical, Data Science, AI/ML, Business Administration (BBA/MBA), Commerce, and Applied Sciences,BA.
- **Cross-Functional Teams:** Teams combining technical developers with domain researchers or UI/UX designers are strongly encouraged to ensure holistic solutions.

3.2 Team Composition Rules

- **Team Size:** Each participating team must consist of a minimum of 3 members and a maximum of 5 members. Solo participation is strictly prohibited under all circumstances.
- **Team Leadership:** Every team must designate one member as the Team Leader, who will act as the single official point of contact for all hackathon correspondence, submissions, and logistical briefings.
- **Inter-College Teams:** Team members may belong to different academic departments or different colleges, provided all members possess valid college identification.
- **No Overlap:** A student can register as a member of only one team. Registering under multiple teams or submitting multiple ideas will result in immediate disqualification of all associated teams.
- **Composition Changes:** Any alteration in team composition following registration must be formally requested and approved in writing by the AdhigamAI Organizing Committee before the Round 1 submission deadline.

3.3 Registration Protocol & Intake Data Fields

Registration is conducted via the official AdhigamAI Hackathon Registration Portal. Teams must provide verified information across the following parameters:

- **1. Team Metadata:** Official Team Name, College/Institution Name, and State.
- **2. Team Leader Profile:** Full Name, College Roll No / Student ID, Academic Branch & Year, Official Email Address, and WhatsApp Contact Number.

- **3. Team Members Roster:** Full Name, Roll No, Branch, Year of Study, Email, and Phone Number for each participating member (2 to 4 additional members).
- **4. Challenge Selection:** Chosen Problem Statement ID (e.g. PS 1, PS 4, PS 7) and Track Domain (Farmers, Army/Defence, Startups, Real-Life, Road Safety).
- **5. Project Summary:** Proposed Project Title, Executive Problem Description (max 150 words), Proposed Solution Architecture (max 200 words), and Key Technologies / AI Stack utilized.
- **6. Round 1 File Upload:** Upload of the official 10-slide PowerPoint presentation in .pptx or .pdf format adhering to the standard naming convention.

4. Official Problem Statements Directory (15 Challenges Across 5 Tracks)

The AdhigamAI Inter-College Hackathon offers 15 precision-crafted problem statements distributed across 5 impact domains. Teams must elect exactly one challenge to address:

ID / Code	Track Domain	Challenge Title	Primary Focus & Technologies
PS 1 (AGRI-01)	FARMERS	Fair Price Discovery for Crops	AI/NLP for multilingual voice/SMS queries (e.g. Hindi, Punjabi, regional dialects), t...
PS 2 (AGRI-02)	FARMERS	Early Warning for Crop Damage	Computer Vision (CV) for crop leaf disease detection via uploaded photos, weather API...
PS 3 (AGRI-03)	FARMERS	Simplified Access to Loans and Government Schemes	Rule-based recommendation engines, conversational AI chatbot in vernacular languages,...
PS 4 (DEF-01)	ARMY / DEFENCE	Health Monitoring for Soldiers in Remote Areas	IoT embedded systems (ESP32, Arduino, Raspberry Pi), low-power Bluetooth (BLE), Edge ...
PS 5 (DEF-02)	ARMY / DEFENCE	Emergency Communication in No-Network Zones	LoRa (Long Range) RF modules, decentralized ad-hoc mesh networking protocols, packet ...
PS 6 (DEF-03)	ARMY / DEFENCE	Community Reporting Near Border Areas	Zero-knowledge proof / cryptographically anonymized metadata, automated geofencing, A...
PS 7 (START-01)	STARTUPS	Local Skilled Worker Marketplace	Geolocation-based matching algorithms, automated algorithmic scheduling, vernacular v...

ID / Code	Track Domain	Challenge Title	Primary Focus & Technologies
PS 8 (START-02)	STARTUPS	Waste-to-Earn Rewards Platform	Computer Vision for waste classification from camera images, QR-code cryptographic ve...
PS 9 (START-03)	STARTUPS	Home-Cooked Meal Delivery Network	Dynamic route optimization for hyper-local delivery batches, demand forecasting to pr...
PS 10 (LIFE-01)	REAL-LIFE PROBLEMS	Reducing Household Water Wastage	Ultrasonic water level sensors, flow-rate pulse meters, microcontrollers (NodeMCU/ESP...
PS 11 (LIFE-02)	REAL-LIFE PROBLEMS	Daily Check-In System for Elderly Living Alone	Conversational AI / IVR voice agents (Twilio, Exotel), automated scheduling cron engi...
PS 12 (LIFE-03)	REAL-LIFE PROBLEMS	Support System for Student Exam Stress	Natural Language Processing (NLP) for empathetic conversational agents, real-time sen...
PS 13 (ROAD-01)	ROAD & VEHICLE SAFETY	Automatic Accident Alert System	Accelerometer & Gyroscope sensor fusion (MPU-6050), edge collision classification alg...
PS 14 (ROAD-02)	ROAD & VEHICLE SAFETY	Helmet and Seatbelt Reminder System	Micro-switch and capacitive head-contact sensors, RF transmitter/receiver link between...
PS 15 (ROAD-03)	ROAD & VEHICLE SAFETY	Pothole and Accident Hotspot Reporting	Smartphone accelerometer Z-axis vibration spike detection, Computer Vision pothole se...

4.1 Comprehensive Problem Statement Profiles

Below are the detailed operational specifications, real-world context, challenge mandates, and expected outcomes for each of the 15 problem statements:

PS 1 Fair Price Discovery for Crops [FARMERS | Code: AGRI-01]

Background & Context	Farmers in rural areas often sell their agricultural produce to local middlemen or agents without knowing the actual prevailing market (mandi) rate. This information asymmetry leads to significant exploitation and substantially lower earnings for the farmer.
The Core Challenge	Design a simple, accessible, low-tech solution (such as a mobile application, SMS broadcast, or Interactive Voice Response / IVR-based phone system) that provides farmers daily verified market prices for their crops across nearby and regional mandis before they negotiate or sell.
Expected Outcome & Deliverables	A functional concept and working prototype (digital app, web dashboard, or IVR flow) demonstrating how a farmer can check fair market prices easily using even a basic feature phone or low-bandwidth smartphone.
Recommended AI / Tech Angle	AI/NLP for multilingual voice/SMS queries (e.g. Hindi, Punjabi, regional dialects), time-series price forecasting, API integration with e-NAM or state agricultural marketing boards, and SMS/WhatsApp gateway integration.

PS 2 Early Warning for Crop Damage [FARMERS | Code: AGRI-02]

Background & Context	Sudden pest infestations, fungal or bacterial plant diseases, and erratic micro-weather shifts (unseasonal frost, hailstorms, heavy downpours) frequently decimate standing crops before farmers can detect the threat or take preventive action, resulting in catastrophic financial loss.
The Core Challenge	Propose an easy-to-use, proactive alert system (via WhatsApp bot, push notifications, localized SMS, or community audio broadcast) that delivers early warnings to farmers regarding pest, disease, and extreme weather risks tailored to their crop type and geographic coordinates.

Background & Context	Sudden pest infestations, fungal or bacterial plant diseases, and erratic micro-weather shifts (unseasonal frost, hailstorms, heavy downpours) frequently decimate standing crops before farmers can detect the threat or take preventive action, resulting in catastrophic financial loss.
Expected Outcome & Deliverables	A clear architectural model, user mockup, and functional alert generation pipeline showing how threat data is collected, processed, and broadcasted to farmers alongside actionable remedy instructions.
Recommended AI / Tech Angle	Computer Vision (CV) for crop leaf disease detection via uploaded photos, weather API ingestion, predictive pest lifecycle modeling based on humidity/temperature curves, and automated vernacular notification dispatch.

PS 3 Simplified Access to Loans and Government Schemes [FARMERS | Code: AGRI-03]

Background & Context	The central and state governments launch numerous agricultural subsidy schemes, crop insurance policies, and low-interest institutional loan facilities. However, smallholder farmers remain largely unaware of their eligibility, and navigating complex government paperwork or multiple physical offices is overwhelming.
The Core Challenge	Create an intuitive mobile application, web portal, or village kiosk interface that evaluates basic farmer demographic and landholding inputs to instantly recommend matching government welfare schemes, subsidies, and credit facilities.
Expected Outcome & Deliverables	A step-by-step interactive application showing how a farmer enters basic criteria (land size, state, crop type, category) and receives a prioritized, jargon-free list of eligible schemes, requisite documents, and application steps.
Recommended AI / Tech Angle	Rule-based recommendation engines, conversational AI chatbot in vernacular languages, optical character recognition (OCR) for Aadhaar/Khasra document pre-filling, and eligibility matching algorithms.

PS 4 Health Monitoring for Soldiers in Remote Areas [ARMY / DEFENCE | Code: DEF-01]

Background & Context	Armed forces personnel deployed at remote border outposts, extreme high-altitude glaciers, or rugged dense terrains face severe physiological hazards including hypothermia, hypoxia (high-altitude pulmonary edema), dehydration, extreme fatigue, and combat injuries that often go undetected until critical.
The Core Challenge	Propose a robust, low-power wearable or kit-integrated biometric monitoring system that continuously tracks vital signs (heart rate, body temperature, blood oxygen saturation/SpO2, physical movement) and automatically alerts field commanders and medical base stations when distress thresholds are breached.

Background & Context	Armed forces personnel deployed at remote border outposts, extreme high-altitude glaciers, or rugged dense terrains face severe physiological hazards including hypothermia, hypoxia (high-altitude pulmonary edema), dehydration, extreme fatigue, and combat injuries that often go undetected until critical.
Expected Outcome & Deliverables	A functional prototype or hardware-software simulation concept showcasing wearable sensor telemetry, real-time threshold analytics, and prioritized alert generation for tactical operations centers.
Recommended AI / Tech Angle	IoT embedded systems (ESP32, Arduino, Raspberry Pi), low-power Bluetooth (BLE), Edge AI anomaly detection models running on low compute, and rugged telemetry visualization dashboards.

PS 5 Emergency Communication in No-Network Zones [ARMY / DEFENCE | Code: DEF-02]

Background & Context	In tactical forward positions, dense jungle warfare, mountainous border frontiers, or disaster-struck zones where commercial cellular towers and internet backbones are non-existent, soldiers are cut off from voice and data networks, creating catastrophic vulnerability during ambushes or medical emergencies.
The Core Challenge	Design an independent, low-cost off-grid tactical communication mechanism (using ad-hoc mesh networking, long-range LoRa RF transceivers, acoustic pulses, or satellite relay simulation) enabling stranded personnel to transmit encrypted SOS coordinates and short emergency telemetry without cellular infrastructure.
Expected Outcome & Deliverables	A working technical model or proof-of-concept demonstrating point-to-point or multi-hop packet transmission of an SOS distress packet across simulated disconnected nodes.
Recommended AI / Tech Angle	LoRa (Long Range) RF modules, decentralized ad-hoc mesh networking protocols, packet encryption (AES-256), peer-to-peer relay algorithms, and lightweight emergency coordinate encoding.

PS 6 Community Reporting Near Border Areas [ARMY / DEFENCE | Code: DEF-03]

Background & Context	Civilian populations living in sensitive border villages frequently observe suspicious vehicular movement, unidentified drones, unauthorized border infiltration, or suspicious packages long before security patrols arrive. However, they lack a safe, swift, and confidential medium to inform defense authorities.
The Core Challenge	Design a secure, tamper-proof, and anonymous reporting portal (mobile app, encrypted web form, automated toll-free SMS/hotline) that empowers local villagers to transmit verified intelligence alerts to military and law-enforcement command centers while safeguarding civilian identity.

Background & Context	Civilian populations living in sensitive border villages frequently observe suspicious vehicular movement, unidentified drones, unauthorized border infiltration, or suspicious packages long before security patrols arrive. However, they lack a safe, swift, and confidential medium to inform defense authorities.
Expected Outcome & Deliverables	A functioning reporting interface featuring geotagged hazard/incident submission, automated identity masking (anonymization), and a military dashboard for prioritizing and corroborating incoming reports.
Recommended AI / Tech Angle	Zero-knowledge proof / cryptographically anonymized metadata, automated geofencing, AI-driven spam/hoax filtering, NLP sentiment and urgency classification, and secure multi-tier officer routing.

PS 7 Local Skilled Worker Marketplace [STARTUPS | Code: START-01]

Background & Context	Informal blue-collar artisans and skilled tradespeople (carpenters, electricians, plumbers, masons, mechanics, tailors) rely almost entirely on irregular word-of-mouth recommendations, leading to severe income volatility. Simultaneously, local households and small businesses struggle to locate reliable, vetted service providers nearby.
The Core Challenge	Design a hyper-local digital marketplace platform (mobile application or progressive web app) that seamlessly connects verified local blue-collar workers with nearby residents and businesses requiring immediate or scheduled trade services.
Expected Outcome & Deliverables	A working platform prototype demonstrating worker onboarding (skill tags, pricing, geo-radius), customer service discovery, booking workflow, and a transparent review/rating trust mechanism.
Recommended AI / Tech Angle	Geolocation-based matching algorithms, automated algorithmic scheduling, vernacular voice-based profile creation for low-literacy workers, and AI-backed customer sentiment analysis for review scoring.

PS 8 Waste-to-Earn Rewards Platform [STARTUPS | Code: START-02]

Background & Context	Civic segregation and recycling initiatives struggle because households, students, and shops have zero tangible incentive to separate dry recyclables (plastic, paper, e-waste, aluminum) from wet municipal garbage, leading to overflowing landfills and wasted circular economy potential.
The Core Challenge	Propose an engaging, gamified platform or point-based reward ecosystem where citizens and commercial establishments earn verifiable digital points, cash-back credits, or merchant discount vouchers for depositing segregated recyclable waste at local civic drop-off hubs.

Background & Context	Civic segregation and recycling initiatives struggle because households, students, and shops have zero tangible incentive to separate dry recyclables (plastic, paper, e-waste, aluminum) from wet municipal garbage, leading to overflowing landfills and wasted circular economy potential.
Expected Outcome & Deliverables	A viable startup business model and working app prototype showcasing user recycling logging, QR/barcode-based deposit verification at collection points, digital wallet reward accrual, and economic sustainability logic.
Recommended AI / Tech Angle	Computer Vision for waste classification from camera images, QR-code cryptographic verification, digital token/reward wallet architecture, and analytics dashboards for recycling partner tie-ups.

PS 9 Home-Cooked Meal Delivery Network [STARTUPS | Code: START-03]

Background & Context	Millions of college students, young professionals, and migrant workers living away from home depend on unhygienic, expensive, or repetitive commercial fast food. Simultaneously, millions of skilled homemakers possess cooking talent and unutilized kitchen capacity but lack the logistics or marketing platform to monetize their skills.
The Core Challenge	Design a decentralized, community-driven startup platform connecting verified home chefs and homemakers cooking fresh, hygienic regional meals with nearby hungry students and working professionals seeking affordable, nutritious tiffin or daily subscriptions.
Expected Outcome & Deliverables	A full-stack product prototype exhibiting the customer ordering and subscription flow, home chef menu publishing and order intake screen, hygiene compliance checklist, and delivery routing framework.
Recommended AI / Tech Angle	Dynamic route optimization for hyper-local delivery batches, demand forecasting to prevent food waste, AI-driven nutritional recommendations, and smart order matching.

PS 10 Reducing Household Water Wastage [REAL-LIFE PROBLEMS | Code: LIFE-01]

Background & Context	Urban and semi-urban localities suffer from chronic groundwater depletion and water scarcity, yet millions of liters of potable treated water are lost daily in residential buildings due to overhead tank overflows, slow-leaking cisterns, faulty taps, and unattended irrigation hoses.
The Core Challenge	Propose a cost-effective, easily retrofittable IoT sensor device or smart monitoring system that detects water leakage or tank overflow in real-time, alerts household members via auditory/mobile notifications, and automatically shuts off supply or prompts behavioral correction.

Background & Context	Urban and semi-urban localities suffer from chronic groundwater depletion and water scarcity, yet millions of liters of potable treated water are lost daily in residential buildings due to overhead tank overflows, slow-leaking cisterns, faulty taps, and unattended irrigation hoses.
Expected Outcome & Deliverables	A working sensor concept or IoT hardware circuit demonstration coupled with an alert dashboard estimating real-time water volume conserved and alerting users before spillage occurs.
Recommended AI / Tech Angle	Ultrasonic water level sensors, flow-rate pulse meters, microcontrollers (NodeMCU/ESP8266), edge logic for anomaly detection (e.g. continuous slow flow at 3 AM indicating a leak), and MQTT cloud push alerts.

PS 11 Daily Check-In System for Elderly Living Alone [REAL-LIFE PROBLEMS | Code: LIFE-02]

Background & Context	An increasing number of senior citizens reside independently while their adult children live in different cities or abroad. In the event of a sudden fall, stroke, cardiac distress, or disorientation, days may pass before anyone notices, often resulting in tragic and preventable outcomes.
The Core Challenge	Design an unobtrusive, zero-stress daily wellness check-in system (such as an automated conversational voice call, a single-tap smartphone widget, or passive motion-sensor heartbeat) that confirms senior safety and automatically triggers multi-tier emergency alerts if the daily check-in is missed.
Expected Outcome & Deliverables	A working prototype demonstrating the automated check-in schedule, conversational confirmation interface (voice or simple button), timeout logic, and multi-channel escalation (SMS/call) to family and neighbors.
Recommended AI / Tech Angle	Conversational AI / IVR voice agents (Twilio, Exotel), automated scheduling cron engines, passive sensor integration (smart plug or PIR motion detection), and fail-safe multi-tier SMS/WhatsApp escalation workflows.

PS 12 Support System for Student Exam Stress [REAL-LIFE PROBLEMS | Code: LIFE-03]

Background & Context	High-stakes competitive examinations, semester finals, and academic performance pressure generate acute anxiety and mental exhaustion among high school and university students. Fear of social stigma, peer competition, and lack of immediate, confidential counseling prevent students from seeking help.
The Core Challenge	Propose an accessible, stigma-free digital emotional support system (mobile application, anonymous peer-to-peer buddy network, or AI-guided cognitive behavioral support agent) that provides real-time stress relief techniques, guided breathing exercises, cognitive reframing, and triage routing to campus counselors.

Background & Context	High-stakes competitive examinations, semester finals, and academic performance pressure generate acute anxiety and mental exhaustion among high school and university students. Fear of social stigma, peer competition, and lack of immediate, confidential counseling prevent students from seeking help.
Expected Outcome & Deliverables	A working interactive application featuring an anonymous user onboarding flow, daily mood/stress logging, interactive de-stressing modules, and an emergency crisis escalation pathway.
Recommended AI / Tech Angle	Natural Language Processing (NLP) for empathetic conversational agents, real-time sentiment and suicide-risk screening algorithms, cognitive behavioral therapy (CBT) conversational frameworks, and privacy-preserving data encryption.

PS 13 Automatic Accident Alert System [ROAD & VEHICLE SAFETY | Code: ROAD-01]

Background & Context	In vehicular collisions and two-wheeler crashes particularly along rural highways or during night commutes critical medical attention (the Golden Hour) is frequently squandered because victims are incapacitated and unable to call for help, while passersby may hesitate to intervene.
The Core Challenge	Design an automated hardware sensor device or smartphone telemetry application that instantaneously detects crash conditions (sudden deceleration, severe impact force, vehicle rollover) and automatically dispatches GPS distress alerts to pre-saved emergency contacts and nearby ambulance centers.
Expected Outcome & Deliverables	A functioning prototype showcasing crash detection logic (via accelerometer/gyroscope sensors or smartphone IMU sensors), an instant countdown cancellation timer (to prevent false alarms), and automatic GPS location broadcast via SMS/Cloud API.
Recommended AI / Tech Angle	Accelerometer & Gyroscope sensor fusion (MPU-6050), edge collision classification algorithms, GPS geofencing, automated emergency SMS dispatch with Google Maps coordinates, and integration with emergency dispatch APIs.

PS 14 Helmet and Seatbelt Reminder System [ROAD & VEHICLE SAFETY | Code: ROAD-02]

Background & Context	A staggering percentage of vehicular fatalities on Indian roads stem directly from failure to buckle seatbelts in cars or wear helmets on motorized two-wheelers. Despite punitive traffic fines, drivers and riders routinely neglect protective gear due to complacency or convenience.
The Core Challenge	Propose an intelligent, low-cost reminder or interlock mechanism (utilizing tactile sensors, infrared/capacitive detection, or Computer Vision) that reliably determines whether a rider is wearing a helmet or driver is buckled up, and enforces compliance via progressive acoustic alerts or engine starter interlocks.

Background & Context	A staggering percentage of vehicular fatalities on Indian roads stem directly from failure to buckle seatbelts in cars or wear helmets on motorized two-wheelers. Despite punitive traffic fines, drivers and riders routinely neglect protective gear due to complacency or convenience.
Expected Outcome & Deliverables	A physical working model or simulated prototype circuit demonstrating headgear/seatbelt engagement detection, state verification, and audible/visual compliance enforcement.
Recommended AI / Tech Angle	Micro-switch and capacitive head-contact sensors, RF transmitter/receiver link between helmet and bike ignition CDI, Computer Vision (YOLO) helmet detection via dashboard camera, and progressive buzzer alert logic.

PS 15 Pothole and Accident Hotspot Reporting [ROAD & VEHICLE SAFETY | Code: ROAD-03]

Background & Context	Unrepaired potholes, washed-out road shoulders, and poorly engineered blind intersections cause thousands of severe road accidents annually. Municipal road authorities often remain unaware of road damage until accidents occur, while citizens find manual complaint filing cumbersome and futile.
The Core Challenge	Design a crowdsourced, automated road safety reporting system (using smartphone sensors mounted on two-wheelers/cars or a citizen reporting app) that captures road distress data with precise geotags, aggregates hotspot density, and delivers an actionable repair dashboard to civic authorities.
Expected Outcome & Deliverables	A working mobile app or IoT prototype demonstrating automated bump/pothole vibration logging (via smartphone accelerometer) or photo-based geotagged reporting, paired with an interactive civic hazard heat map.
Recommended AI / Tech Angle	Smartphone accelerometer Z-axis vibration spike detection, Computer Vision pothole segmentation on camera footage, GIS spatial clustering algorithms (DBSCAN) to consolidate duplicate reports, and municipal priority ranking dashboards.

5. Round 1 Online Idea Qualifier & PPT Submission Guidelines

Round 1 serves as the competitive screening filter. Juries will review all eligible submissions to identify top-tier projects that demonstrate genuine innovation, technological feasibility, and a clear path toward a working prototype.

5.1 Technical Submission Rules

- **Allowed Formats:** Submissions must be uploaded as either Microsoft PowerPoint (.pptx) or Adobe Acrobat PDF (.pdf). Link-only presentations (e.g. Canva, Google Slides links) must be exported to PDF/PPTX prior to upload.
- **Strict Slide Limit:** The presentation deck must contain 8 to 10 content slides maximum, excluding the title slide. Decks exceeding 10 content slides will face automatic scoring penalties.
- **File Size Limit:** Maximum allowable file size is 20 MB. Compress embedded high-resolution graphics if necessary.

- **Mandatory Naming Convention:** The file must be named precisely as follows: TeamName_TrackName_PSCode_R1.pptx (Example: TechTitans_Farmers_AGRIO1_R1.pptx).
- **Legibility & Visual Quality:** Body typography must maintain a minimum readable size of 18pt. Limit visual design to no more than 2 font families per slide. Diagrams and flowcharts must be crisp and legible.

5.2 Required 10-Slide Deck Blueprint

To ensure fair and uniform benchmarking, all participating teams must structure their Round 1 presentation deck adhering to the exact 10-slide sequence outlined below:

Slide #	Slide Section Title	Mandatory Contents, Diagrams & Metrics to Include
Slide 1	Title Slide & Team Roster	Official Team Name, College/Institution name with city, Chosen Track & Problem Statement ID/Code, Project Title, and Team Roster (Leader & Member full names, student IDs, and departments).
Slide 2	Problem Statement & Context Analysis	Sharp articulation of the specific pain point addressed; target user persona; quantifiable scale of the problem in India; evidence of why existing conventional methods fail.
Slide 3	Proposed Solution Concept	High-level overview of your proposed solution in intuitive terms; core value proposition; step-by-step user journey or operational workflow explaining how it directly eliminates the root problem.
Slide 4	Innovation & Uniqueness (USP)	Clear explanation of what sets your idea apart from existing commercial or open-source solutions; Unique Selling Proposition (USP); technical, behavioral, or economic novelty.
Slide 5	AI & Technical Architecture	Complete architectural block diagram illustrating data pipelines, AI/ML models (e.g. computer vision, NLP, regression), cloud services, hardware/sensor components, APIs, and frameworks.
Slide 6	Technical Feasibility & Implementation Logic	Evaluation of data availability, latency constraints, edge vs cloud compute feasibility, offline accessibility, hardware retrofittability, and risk mitigation strategies.
Slide 7	Scalability, Social & Business Impact	Quantifiable metrics of success (e.g. estimated liters of water saved, hours saved for farmers, response time reduction); unit economics or sustainability model; future expansion potential.
Slide 8	Implementation Plan & Build Milestones	Chronological development roadmap detailing milestones achieved to date, current development stage, and the concrete technical build planned for the Round 2 Offline Finale.
Slide 9	Prototype Snapshot / Wireframe / Schema	Visual evidence of progress: screenshots of working UI, clickable Figma wireframes, circuit schematic diagrams, hardware 3D mockups, API endpoint tests, or code repository snippets.

Slide #	Slide Section Title	Mandatory Contents, Diagrams & Metrics to Include
Slide 10	Team Roles, Future Vision & Conclusion	Individual team member roles and specific contributions to the build; summary of key project takeaways; future milestones post-hackathon; formal concluding statement.

6. Round 2 Offline Pitch Round & Live Prototype Finale

The Grand Finale of the AdhigamAI Inter-College Hackathon is conducted physically at the AdhigamAI Master Hub located at Sri Sukhmani College of Technology & Management, Dera Bassi, Punjab. This offline round brings together the most promising innovator teams to pitch their solutions directly before a high-caliber jury comprising industry AI architects, venture mentors, and senior faculty.

6.1 Grand Finale Day Schedule (Run-of-Show)

The offline finale is organized as an intensive, single-day pitch event adhering to the official schedule below:

Time Slot	Activity / Stage	Purpose & Focus	Participant Team Action	Jury & Mentor Action
09:00 - 10:00	Check-In & Verification	Physical verification of teams, registration confirmation, and workspace allocation.	Report to Registration Desk, verify College IDs, collect official team credentials.	Registration Desk staff issues booth assignments and name badges.
10:00 - 10:30	Opening & Briefing	Official welcome, orientation, pitch format alignment, and introduction of jury.	Assemble in Main Auditorium; receive final pitch instructions and running order.	Introduction of Jury Panel and explanation of scoring criteria.
10:30 - 11:15	Prototype & Booth Setup	Technical calibration, local server deployment, display testing, and hardware prep.	Set up prototype hardware/software at allocated station; test projector connections.	Technical mentors available for facility assistance; judges pre-briefing.
11:15 - 13:15	Pitch & Demo: Session 1	Evaluation of the first batch of finalist teams (Morning Pitch Batch).	Deliver formal PPT pitch (5-6 min) + live working demo (3-4 min) + answer jury Q&A (2-3 min).	Jury scores each pitching team across Round 2 rubric parameters.
13:15 - 14:00	Lunch & Networking Break	Participant rejuvenation and peer-to-peer networking.	Enjoy provided lunch and network with fellow innovators and mentors.	Judges lunch break and informal check-in on Session 1 scoring trends.

Time Slot	Activity / Stage	Purpose & Focus	Participant Team Action	Jury & Mentor Action
14:00 - 16:00	Pitch & Demo: Session 2	Evaluation of remaining finalist teams (Afternoon Pitch Batch).	Deliver formal PPT pitch (5-6 min) + live working demo (3-4 min) + answer jury Q&A (2-3 min).	Jury scores afternoon teams across Round 2 rubric parameters.
16:00 - 16:30	Closed Jury Deliberation	Compilation of all pitch scores, consensus normalization, and winner selection.	Free networking, booth interaction, and informal demonstrations to peers.	Jury compiles all scorecards, resolves ties, and finalizes podium rankings.
16:30 - 17:00	Valedictory & Awards	Public felicitation, presentation of trophies, certificates, and closing address.	Attend Valedictory Ceremony; receive awards, certificates, and participate in group photo.	Jury feedback address, announcement of winners, and felicitation.

6.2 Strict Pitch Slot Timing Protocol (10-12 Minutes Total)

Time management is rigorously monitored by official timekeepers. Each team is allocated a strict 10 to 12 minute evaluation slot structured as follows:

- **Segment 1: Formal Slide Presentation (5-6 Minutes):** The team delivers their executive pitch using their refined PowerPoint deck. Teams must explain the core problem, their value proposition, AI/engineering architecture, market viability, and team capability. A warning bell will ring at the 5-minute mark.
- **Segment 2: Live Working Prototype Demonstration (3-4 Minutes):** The team transitions immediately into demonstrating their live functional prototype, IoT device, mobile app, or interactive interface. Judges must see the software running live, processing input, or the hardware executing sensor telemetry. Mockup videos or screenshots are permitted only as supplementary aids.
- **Segment 3: Interactive Jury Q&A & Technical Defense (2-3 Minutes):** The jury panel asks probing questions regarding code implementation, AI model selection, dataset validity, edge latency, cost feasibility, and commercial scalability. All team members must be prepared to speak.

6.3 Prototype Guidelines & Maturity Tiers

Because the offline finale is a dedicated Pitch Round, finalists must arrive with an advanced, functional prototype ready to be demonstrated. Prototypes will fall into one of three tiers:

- **Tier 1: Functional Software / Web / Mobile App MVP:** Working application with live backend APIs, database connectivity, and functional UI demonstrating core user workflows and AI inference.
- **Tier 2: Hardware / IoT / Embedded Circuit Prototype:** Physical microcontroller (Arduino, ESP32, Raspberry Pi) integrated with active sensors, telemetry transmitters, or mechanical actuators demonstrating live signal transmission and alert triggers.
- **Tier 3: Clickable Interactive High-Fidelity Prototype:** Fully interactive, navigable Figma or proto.io simulations demonstrating dynamic user interactions, designed specifically for complex systems where backend deployment requires restricted institutional access.

7. Official Evaluation Criteria & Scoring Rubrics

Evaluation is conducted with complete objectivity across both rounds utilizing the exact mathematical criteria and weightages established in the AdhigamAI Hackathon Structure workbook:

7.1 Round 1 Online Idea Evaluation Rubric (100% Scale)

Evaluation Parameter	Weightage	Key Evaluation Criteria & Jury Benchmarks	Score Range
Problem Relevance	15% (0.15)	Clarity of problem articulation, acute understanding of user pain points, and societal/industry significance.	0 15 pts
Innovation & Originality	25% (0.25)	Novelty of the concept, differentiation from existing market products, and intellectual creativity.	0 25 pts
AI / Technology Approach	20% (0.20)	Soundness of the technical architecture, appropriateness and depth of AI/ML integration, and data logic.	0 20 pts
Feasibility & Build Logic	20% (0.20)	Practical viability of engineering the solution; realistic resource requirements, constraints, and timeline.	0 20 pts
Expected Impact	10% (0.10)	Potential quantifiable value, community empowerment, economic return, and scalability of the solution.	0 10 pts
Presentation Quality	10% (0.10)	Visual structure, readability, adherence to the 10-slide blueprint, concise articulation, and data completeness.	0 10 pts

7.2 Round 2 Offline Pitch & Prototype Evaluation Rubric (100% Scale)

Evaluation Parameter	Weightage	Key Evaluation Criteria & Jury Benchmarks	Score Range
Problem Solution Fit	10% (0.10)	How directly, elegantly, and completely the solution eliminates the stated challenge.	0 10 pts
Innovation & Creativity	15% (0.15)	Breakthrough features, unique design thinking, and intellectual originality demonstrated in practice.	0 15 pts
AI / Technical Implementation	20% (0.20)	Depth of technical execution, code quality, algorithm robustness, and effective use of modern AI models.	0 20 pts
Working Prototype	25% (0.25)	Completeness, functionality, real-time stability, and demo-readiness of the software/hardware prototype.	0 25 pts

Evaluation Parameter	Weightage	Key Evaluation Criteria & Jury Benchmarks	Score Range
Feasibility & Practicality	10% (0.10)	Practical deployability in low-resource environments, manufacturing cost, and operational maintenance.	0 - 10 pts
Impact & Scalability	10% (0.10)	Quantifiable societal benefit, commercial viability, and pathway for scaling to thousands of end users.	0 - 10 pts
Presentation, Demo & Q&A	10% (0.10)	Clarity of verbal pitch, seamless live demo flow, confidence, and technical mastery during jury Q&A.	0 - 10 pts

8. Code of Conduct, AI Policy & Disqualification Matrix

8.1 Code of Conduct & Academic Integrity

- **Original Intellectual Property:** All submitted ideas, architectural designs, algorithms, and prototype code must represent the original work of the registered team members.
- **Open Source Usage:** Teams are permitted to utilize open-source libraries, publicly accessible APIs, pre-trained machine learning models (e.g. Hugging Face, YOLO, OpenCV), and standard software development kits (SDKs), provided all dependencies are explicitly acknowledged and attributed in the documentation.
- **Decorum & Inclusivity:** Participants must conduct themselves with absolute professional courtesy and respect toward fellow contestants, student coordinators, technical volunteers, institutional hosts, and jury members.

8.2 Policy on Generative AI & Developer Copilots

AdhigamAI recognizes Generative AI as an indispensable modern developer multiplier. However, its use is governed by clear boundaries:

- **Allowed Assistance:** Teams may use Generative AI tools (such as ChatGPT, Claude, GitHub Copilot, Gemini) for ideation assistance, architectural research, boilerplate coding, regex formulation, and debugging support.
- **Mandatory Disclosure:** If Generative AI tools were used substantially in designing architecture or generating code modules, teams must disclose this in Slide 5 of their presentation deck.
- **Technical Ownership:** AI tools must never replace fundamental technical comprehension. If a team submits AI-generated code or architectural components that they cannot explain or defend during jury questioning, the team will receive an immediate score of zero for technical implementation.

8.3 Grounds for Immediate Disqualification

Violation of any of the following mandatory conditions will trigger instant disqualification from the competition without recourse to appeal:

- **1. Plagiarism & IP Theft:** Direct copying of existing commercial products, academic research papers, or fellow participants' submissions without significant original technical contribution.
- **2. Submission Deadline Violation:** Uploading decks or project files after the official submission deadline has elapsed.
- **3. Format & Structure Non-Compliance:** Gross deviation from the mandatory 10-slide structure, exceeding the slide limit substantially, or submitting locked/corrupted files.
- **4. Misrepresentation of Prototype Capabilities:** Falsifying prototype functionality, presenting pre-recorded commercial product demos as one's own code, or faking live hardware sensor data.

- **5. Dual Registration:** Any individual student appearing as a registered member of more than one team.
- **6. No-Show at Finale:** Failure of registered team members to report physically to the venue for the Round 2 Pitch Round without prior written exemption.
- **7. Harassment & Misconduct:** Any act of harassment, offensive language, intentional damage to campus property, or tampering with rival teams' demo stations.

9. Prizes, Recognition, Incubation & Career Opportunities

The AdhigamAI Inter-College Hackathon offers substantial tangible rewards, prestigious academic citations, and long-term career advancement pathways:

Award Category / Title	Prizes & Honors Conferred	Incubation, Mentorship & Career Benefits
Grand Winner (1st Place)	Cash Award / Innovation Grant + Grand Championship Trophy + Certificate of Excellence	Direct fast-track entry into the AdhigamAI Startup Incubation Cohort; dedicated industry mentorship; investor demo day access.
First Runner-Up (2nd Place)	Cash Award + Official Runner-Up Trophy + Certificate of Excellence	Incubation support, prototype refinement grant consideration, and cloud compute credits.
Second Runner-Up (3rd Place)	Cash Award + Podium Trophy + Certificate of Excellence	Mentorship access, technical project acceleration, and certificate of high merit.
All Shortlisted Finalists & Participants	Official Certificate of Finalist Distinction / Certificate of Participation	Eligibility to apply for AdhigamAI Campus Chapter Leadership (Head, Tech Head, Ambassador) and direct internship interview shortlisting.

9.1 Post-Hackathon Career & Incubation Fast-Track

Beyond one-time prize distribution, high-performing participants become integral members of the broader AdhigamAI innovation network:

- **AdhigamAI Club Induction:** High-performing participants are prioritized for selection as Committee Heads, Sub-Heads, Tech Heads, and Campus Ambassadors for the official AdhigamAI College Club chapters at their institutions.
- **Industry Internships:** Outstanding software and hardware builders are shortlisted for research fellowships, technical internships, and client AI implementation roles with AdhigamAI partner organizations.
- **Full IP Retention:** Participating teams retain 100% intellectual property (IP) ownership of all source code, algorithms, and designs developed during the hackathon. AdhigamAI retains non-exclusive rights to showcase project highlights, media, and video recordings for educational and publicity purposes.

10. Frequently Asked Questions (FAQ)

Q: Can 1st-year students compete against final-year students ?

Yes. The AdhigamAI Hackathon evaluates the ingenuity, feasibility, and technical execution of the project rather than academic seniority. 1st-year teams with sharp problem understanding and clean execution regularly score high on innovation and prototype delivery.

Q: Can students from different departments or colleges form a combined team

Yes. Inter-disciplinary teams combining students from Computer Science, Electronics, Mechanical, and Business or Design are actively encouraged. Cross-college teams are also permitted as long as all members carry valid college IDs.

Q: Is there any registration fee to participate in the hackathon

No. Participation in the AdhigamAI Inter-College Hackathon is completely free of charge. No registration fees or submission charges are collected at any stage.

Q: What is the exact difference between Round 1 and Round 2

Round 1 is a remote online screening phase where teams submit an exhaustive 10-slide PowerPoint deck explaining their proposed solution, AI architecture, and build plan. Round 2 is an offline finale where shortlisted teams travel to Sri Sukhmani College to pitch in person, demonstrate a working prototype live on stage, and face jury Q&A.

Q: Do we have to code our solution from scratch at the offline venue

No. Round 2 is an Offline Pitch Round, not an overnight coding hackathon. Finalists build, test, and refine their prototypes prior to arriving at the venue. At the venue, teams set up their demo stations, pitch via PPT, demonstrate their live working prototype, and answer jury questions.

Q: What happens if our prototype hardware fails during the live demo

Judges evaluate engineering progress, system architecture, and technical direction. While a live working demo carries maximum points, teams should always carry pre-recorded screencasts or fallback video clips of successful bench tests in case of unexpected on-stage hardware glitches.

Q: Can we modify our PPT between Round 1 and Round 2

Yes. Finalists are permitted and encouraged to refine their PPT to reflect progress made on their prototype, updated UI screenshots, and refined performance metrics. However, the core Problem Statement, project title, and 10-slide structural order must remain consistent.

Q: Are we allowed to use pre-trained AI models or third-party APIs

Yes. Teams may leverage pre-trained models (e.g. YOLO, MobileNet, BERT, Whisper) and third-party APIs (e.g. Twilio, Google Maps, OpenWeatherMap) provided they clearly articulate how these models are integrated into their proprietary architecture and disclose dependencies in Slide 5.

Q: Who owns the intellectual property (IP) of the submitted project

The participating team retains 100% ownership of all intellectual property, patent rights, code, and trade secrets developed. AdhigamAI and host institutions claim zero commercial ownership.

Q: Will travel allowance (TA) or accommodation be provided for Round 2

Detailed logistical guidelines regarding local hospitality, hostel accommodation, or campus dining passes are provided in the Finalist Confirmation Package sent to shortlisted teams following Round 1 results.

11. Organizing Committee & Support Directory

For any inquiries, clarifications, or technical support regarding registration, problem statements, or finale logistics, participants may contact the official organizing secretariat:

MAMTA GAUR
+91 8699011982
mamta.gaur@focalyt.com

GRAND FINALE PHYSICAL VENUE & HELPDESK DETAILS

Host Institution: Sri Sukhmani College of Technology & Management (SSCOTM), AdhigamAI Master Hub.
Address: Chandigarh-Ambala Highway, Dera Bassi, District SAS Nagar (Mohali), Punjab 140507, India.
Official Portal: <https://adhigamai.com/hackathon2026> | Helpdesk Hours: 09:00 AM - 06:00 PM IST (Mon-Sat).

We wish all participating innovators the very best of success! Build with purpose, engineer with passion, and create lasting real-world impact.